

Directions for use Read carefully!

Ringer's Injection

Full electrolyte solution for replacement of extracellular fluid

Composition

Each 100 ml contains:

Sodium Chloride	0.860 g
Potassium Chloride	0.030 g
Calcium Chloride, 2H ₂ O	0.033 g
Water for Injections to	100 ml

Electrolytes:	mmol/l (mEq/l)
Na ⁺	147.0
K ⁺	4.0
Ca ⁺⁺	2.3 (4.5)
Cl ⁻	155.5
Osmolarity:	309 mOsm/l

Characteristics

The electrolyte content of Ringer's Injection resembles that of extracellular fluid. The excess of about 50 mmol/l chloride (as compared to extracellular fluid) counteracts alkalotic tendencies.

Indications

Extracellular fluid loss
Sodium and chloride depletion
Light metabolic alkalosis

Dosage

Average dose : 1000 ml/day.

Drip rate : 120 - 180 drops/min corresponding to 360 - 540 ml/h.

Route of administration I.V.

Contraindications

Hypokalemia
Hypernatraemia
Hyperchloremia
Metabolic acidosis
Renal insufficiency

caution in patients with hypertension. Blood glucose, serum electrolytes and water balance should be monitored regularly.

Stability of any additives to this solution should be checked before use.

Effects

When the body cannot adequately utilize or excrete any particular ion, it may accumulate to give rise to characteristic or elevated levels of that particular ion.

Use with
care and
monitor

The comp

Side effects

In the event
of a symptom

Symptoms and treatment for overdosage

Since the concentration of ions in this preparation mimics normal plasma levels, it is unlikely to cause ionic imbalance to any great extent. Any such tendency should be readily detected in the routine serum electrolyte monitoring.

However, the development of any of the following symptoms will require close scrutiny of blood electrolyte levels and appropriate management:-

- Nausea, vomiting, diarrhoea, constipation, anorexia
- Abdominal pain, abdominal cramps
- Listlessness, weakness (general or muscular)
- Restlessness
- Thirst, dry mouth, swollen tongue, polyuria
- Pyrexia
- Paralysis
- Bone pain
- Dizziness, drowsiness, confusion
- Cardiac complications

If the serum electrolyte check shows any particular ion to be elevated, it should then be managed accordingly.

General Management of:-

1. **Hypernatraemia**
 - reduce/eliminate Na⁺ intake
 - in severe cases, treat with dialysis
2. **Hyperkalaemia**
 - eliminate K⁺ administration
 - withdraw any concomitant K⁺ sparing diuretics
 - serum K⁺ levels may be brought down by infusion of up to 50 - 125 grams of glucose over 1 hour with insulin
 - severe cardiac toxicity, requiring immediate attention, may be treated with 10 - 20 ml of 10% Calcium Gluconate (IV) over 1 - 5 mins
 - severe cases may also benefit from dialysis
3. **Hypercalcaemia**
 - eliminate Ca⁺⁺ administration
 - increase fluid intake to combat dehydration and deposition of Calcium in the kidneys
 - administer Sodium Phosphate PO or IV to reduce serum calcium

Usage during pregnancy

No adverse reaction has been reported.

Shelf life

The product must not be used beyond the expiry date stated on the label.

Storage

The product should not be stored above the temperature stated on the label.

Presentation

500 ml or 1000 ml plastic container.
500 ml or 1000 ml PVC bags.

Method of administration

In the special case of rapid infusion under external pressure which may be necessary in emergency situations, before starting the infusion, all air must be removed from containers with air space inside, as otherwise there is a risk of producing air embolism during the infusion.

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Manufactured by:

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